Demystifying Linear Regression

Dr. Kevin Mackay
Data Analyst at Statistics Canada

Why Use Linear Regression?

- Linear regression is a mathematical model used to:
  1. **Understand relationships** between the dependent variable (y) and one or more independent variables (x)
     - How does the age of a vehicle (x) relate to its likelihood of being involved in a traffic accident (y)?
  2. **Predict the dependent variable** (y) based on one or more independent variables (x)
     - How much does CO2 emissions (y) decrease for every additional kilometer cycled per capita (x) in a city?

Equation Of A Line

$$y = mx + b$$

$$y = ax + b$$

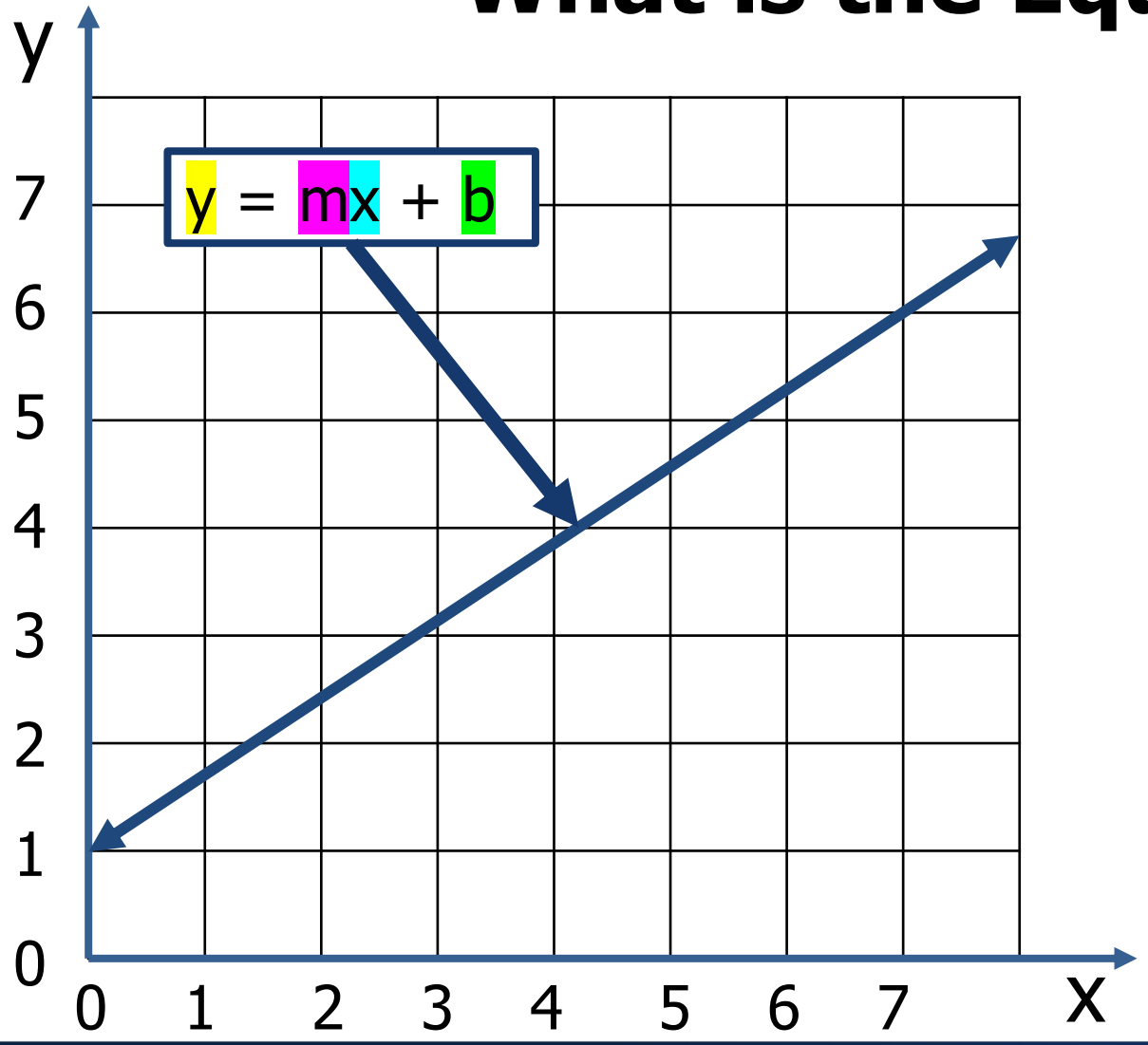
$$y = ab + c$$

Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1$$

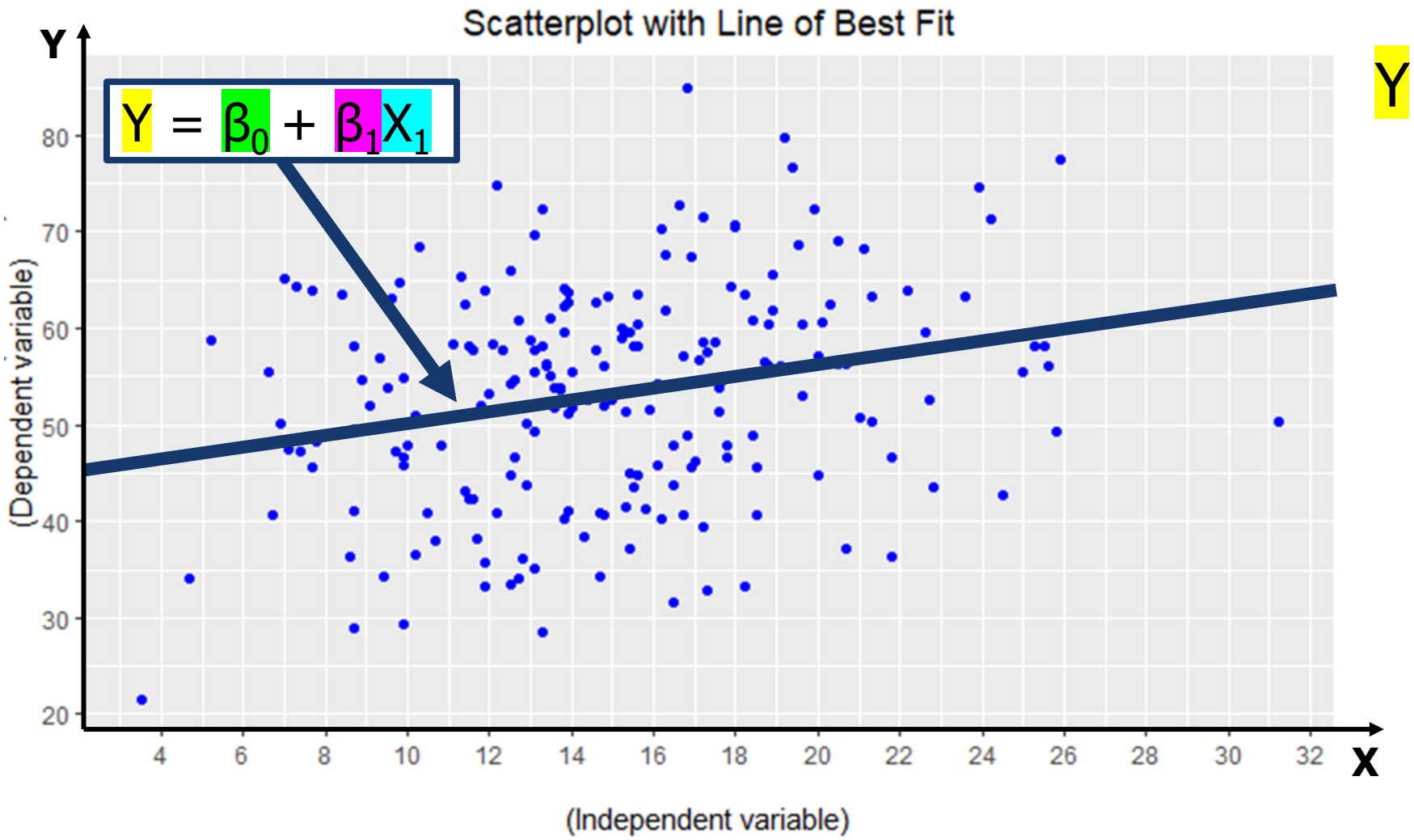
- **Y**: the value of your dependent variable
- **β_0**: y-intercept, value of y when x = 0
- **β_1**: slope for predictor variable X_1
- **X_1**: the value of your independent variable

What is the Equation of a Line?



- y = mx + b
  - y: how far up
  - x: how far along
  - m: slope or gradient
  - b: value of y when x = 0
    - Also known as the y-intercept

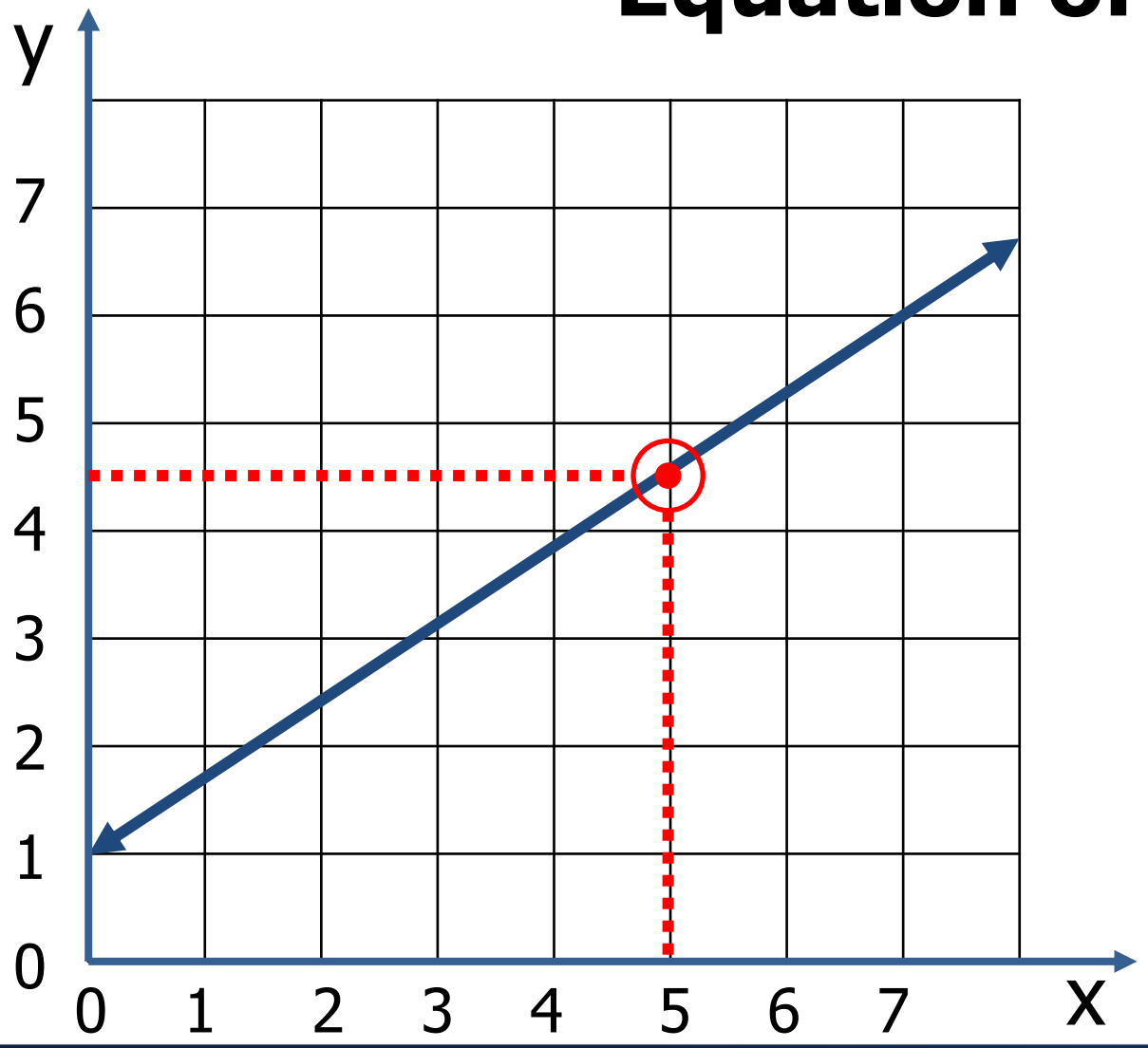
What is a Linear Regression Model?



$Y = \beta_0 + \beta_1 X_1$

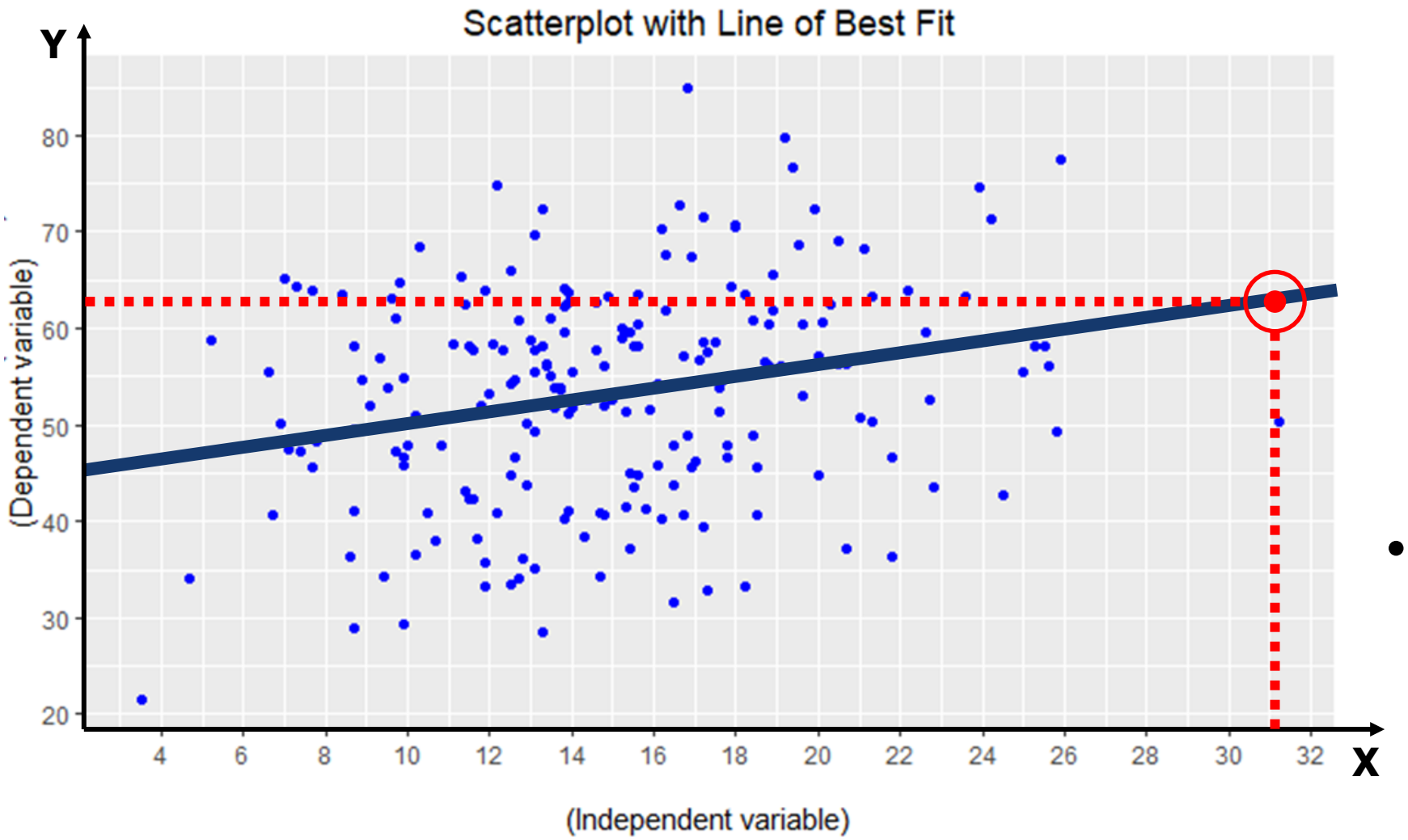
- Y: how far up
- X_1: how far along
- β_1: slope or gradient
  β_0: value of Y when $X_1 = 0$
  - Also known as the y-intercept

Equation of a Line: y & x



- y = mx + b
  - y: how far up
  - x: how far along
- Example
  - x = 5
  - y = 4.5

Linear Regression Model: Y & X_1

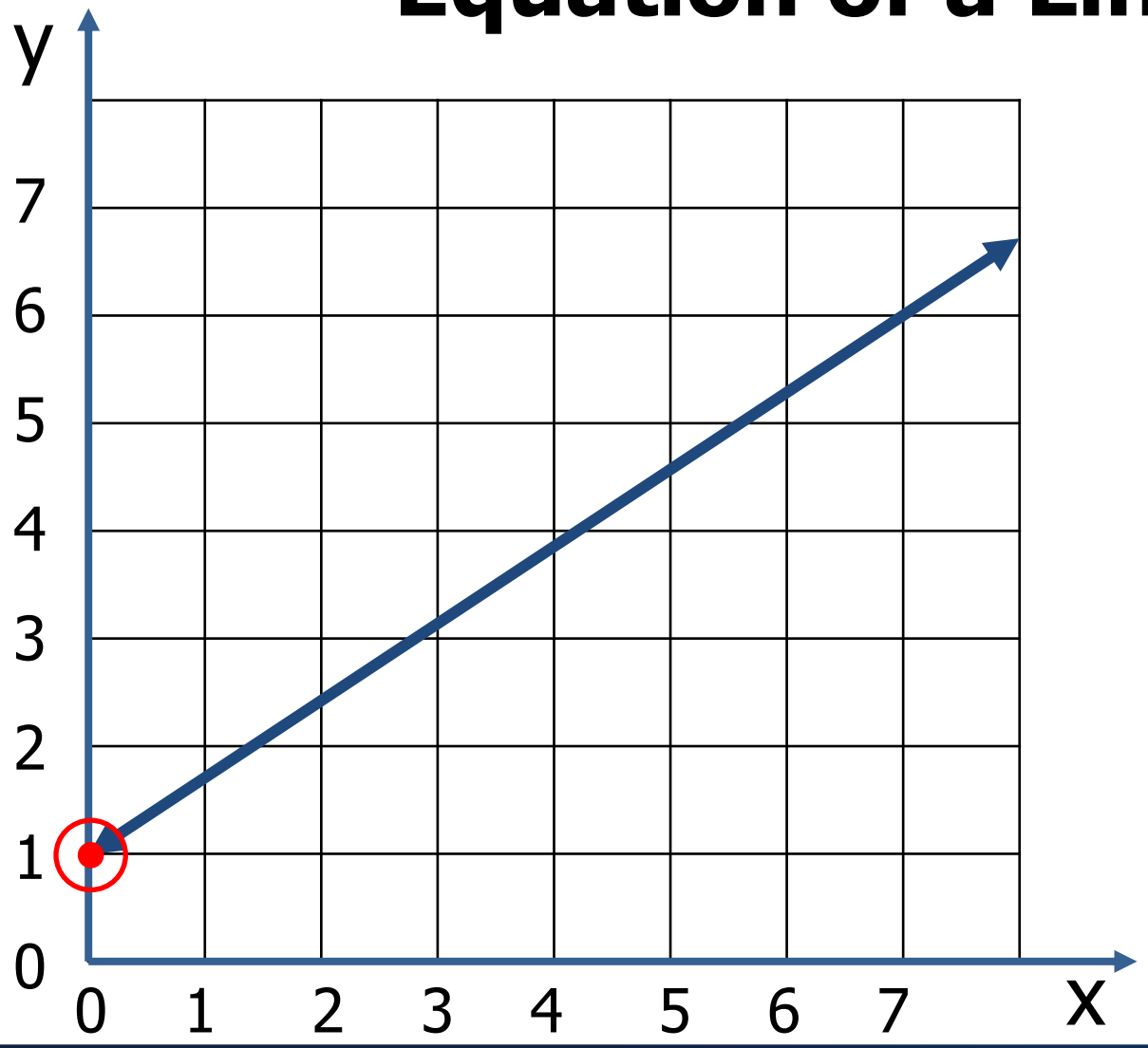


$Y = \beta_0 + \beta_1 X_1$

- Y: value of dependent variable
- X_1: value of independent variable

- Example
  - $X_1 = 31$
  - $Y = 63$

Equation of a Line: b (y-intercept)

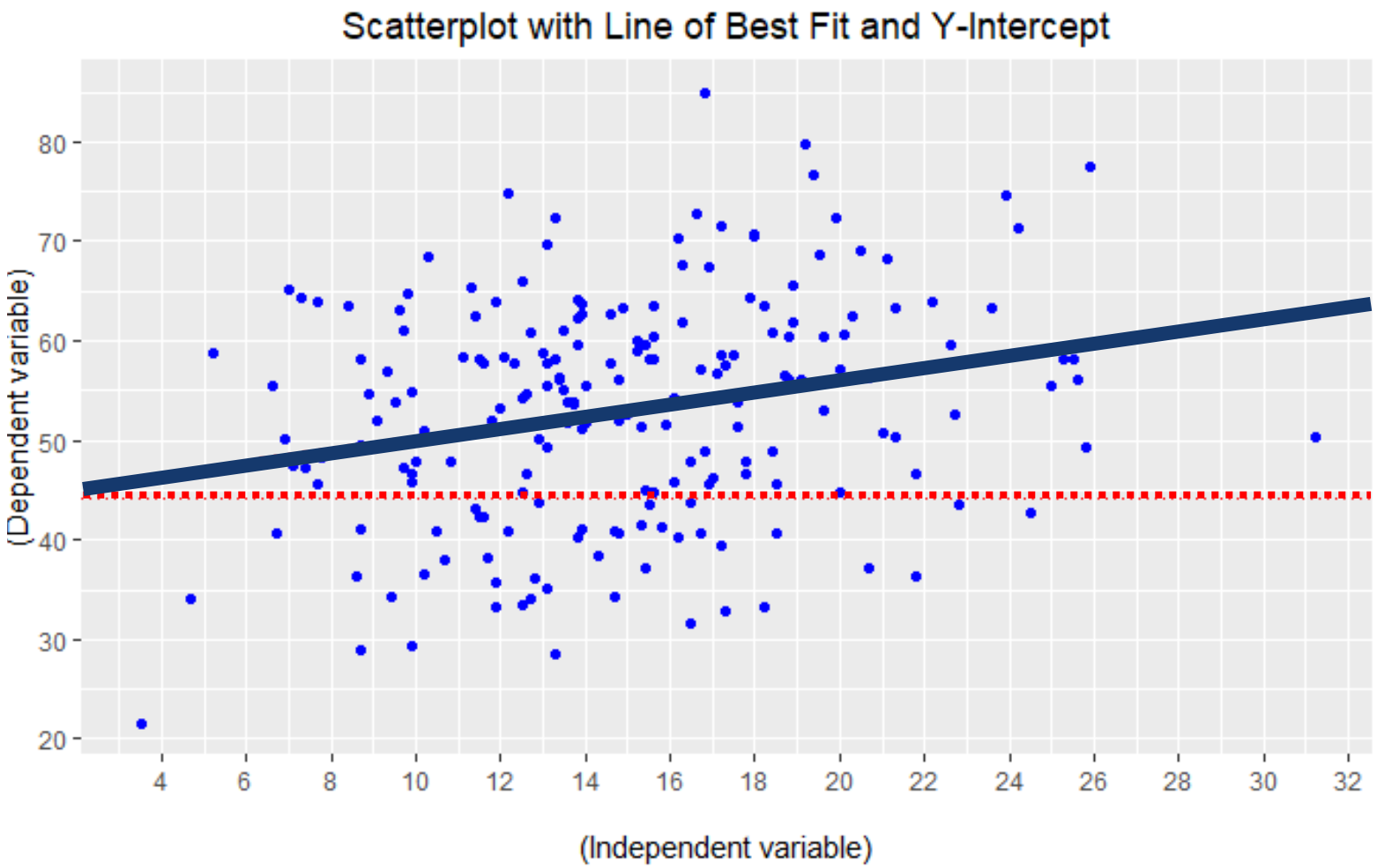


- $y = mx + b$
  - b: value of y when $x = 0$
    - Also known as the y-intercept

$b = 1$

When $x = 0$, $y = 1$

Linear Regression Model: β_0 (Y-intercept)



$Y = \beta_0 + \beta_1 X_1$

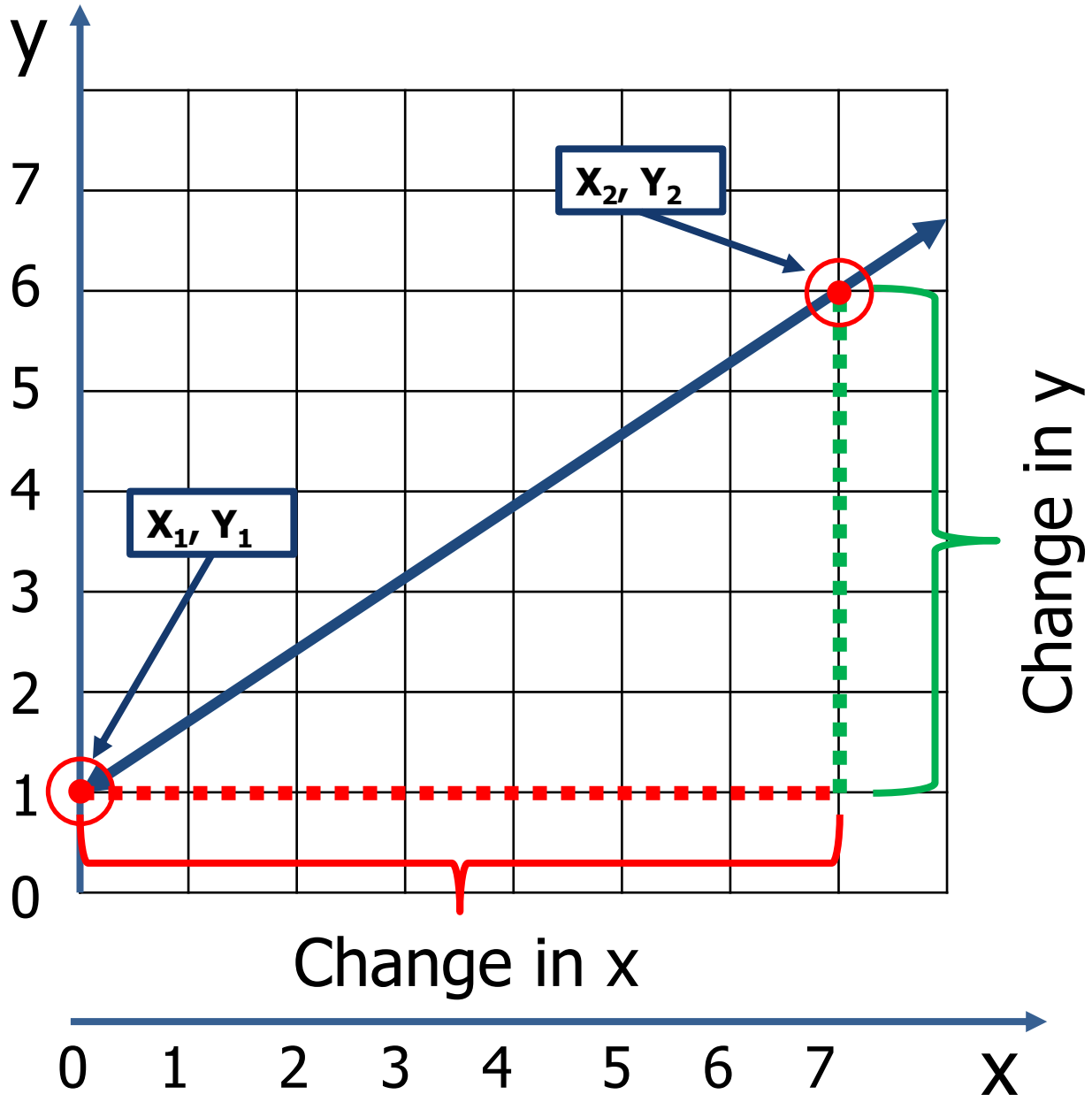
- β_0: the value of the dependent variable when the independent variable is zero

Example

- $\beta_0 = 44.2$

If the independent variable is 0, dependent variable equals 44.2

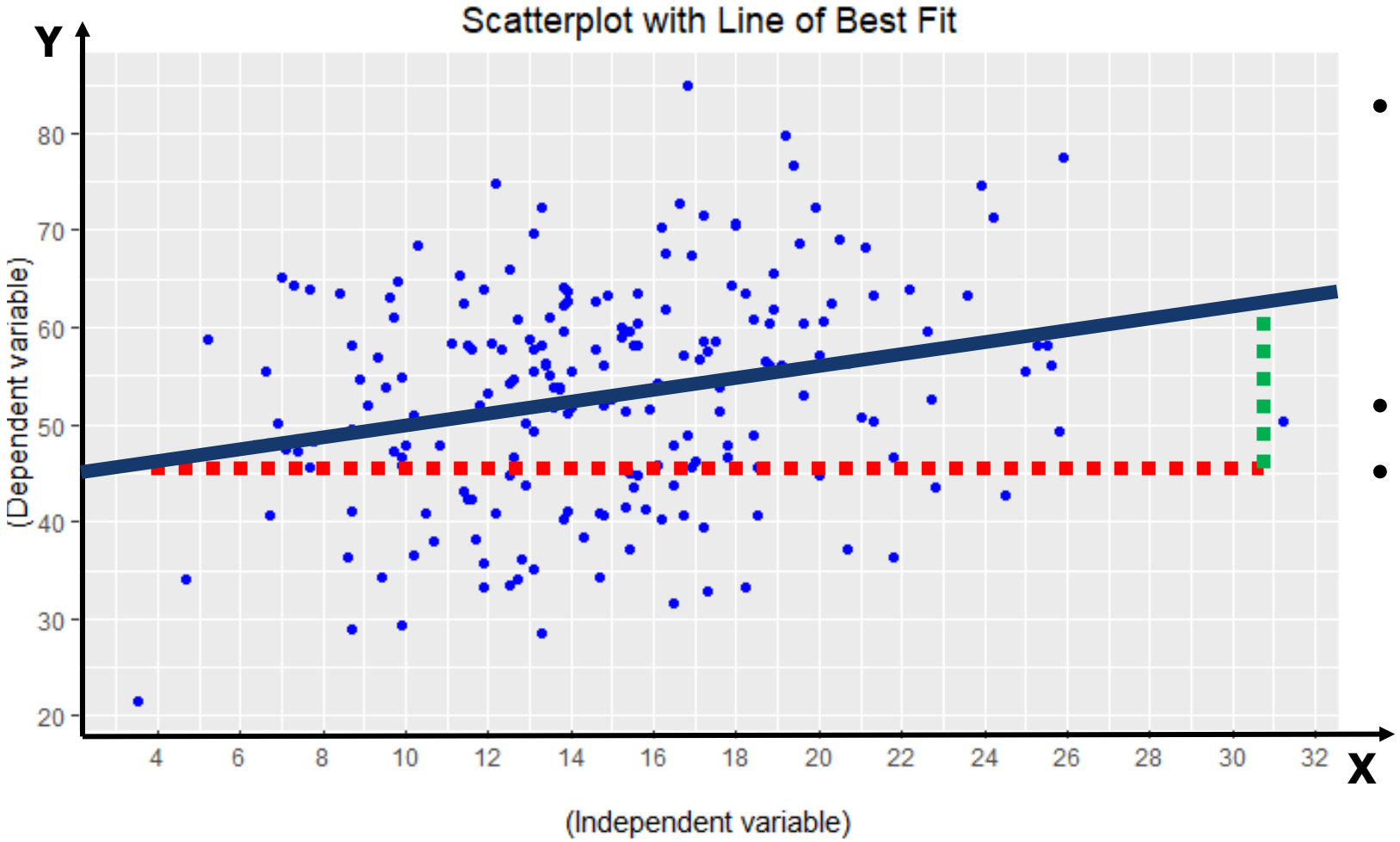
Equation Of A Line: m (slope)



- m = rise / run
- = change in y / change in x
- = $(y_2 - y_1) / (x_2 - x_1)$
- = $(6 - 1) / (7 - 0)$
- = 5 / 7
- m = 0.71

- For every 1 unit increase in x, y increases by 0.71

Linear Regression Model: β_1 (relationship between X_1 and Y)



- $Y = \beta_0 + \beta_1 X_1$
  - β_1: Slope of the regression line, or the relationship between X_1 and Y

- $\beta_1 = 0.61$
- For every 1 unit increase in the independent variable, the dependent variable will increase by 0.61

Predicting y & Y

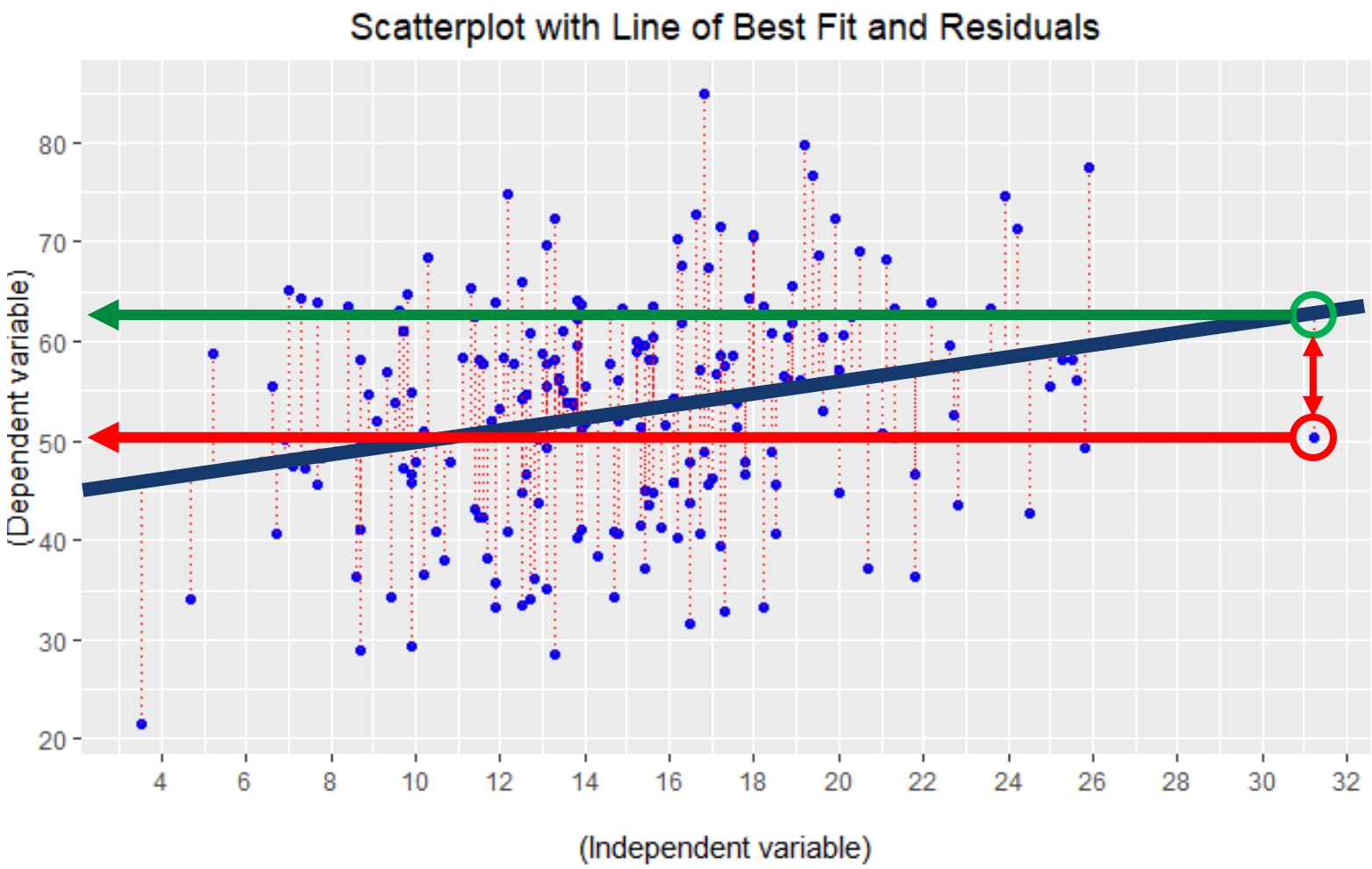
- $y = mx + b$
- $y = (0.71)(x) + 1$

- What is the value of y when $x = 25$?
- $y = (0.71)(25) + 1$
- $y = 18.75$
- If $x = 25$, $y = 18.75$

- $Y = \beta_0 + \beta_1 X_1$
- $Y = 44.2 + (0.61)(X_1)$

- What is the value of Y when $X_1 = 25$?
- $Y = 44.2 + (0.61)(25)$
- $Y = 59.45$
- If $X_1 = 25$, $y = 59.45$

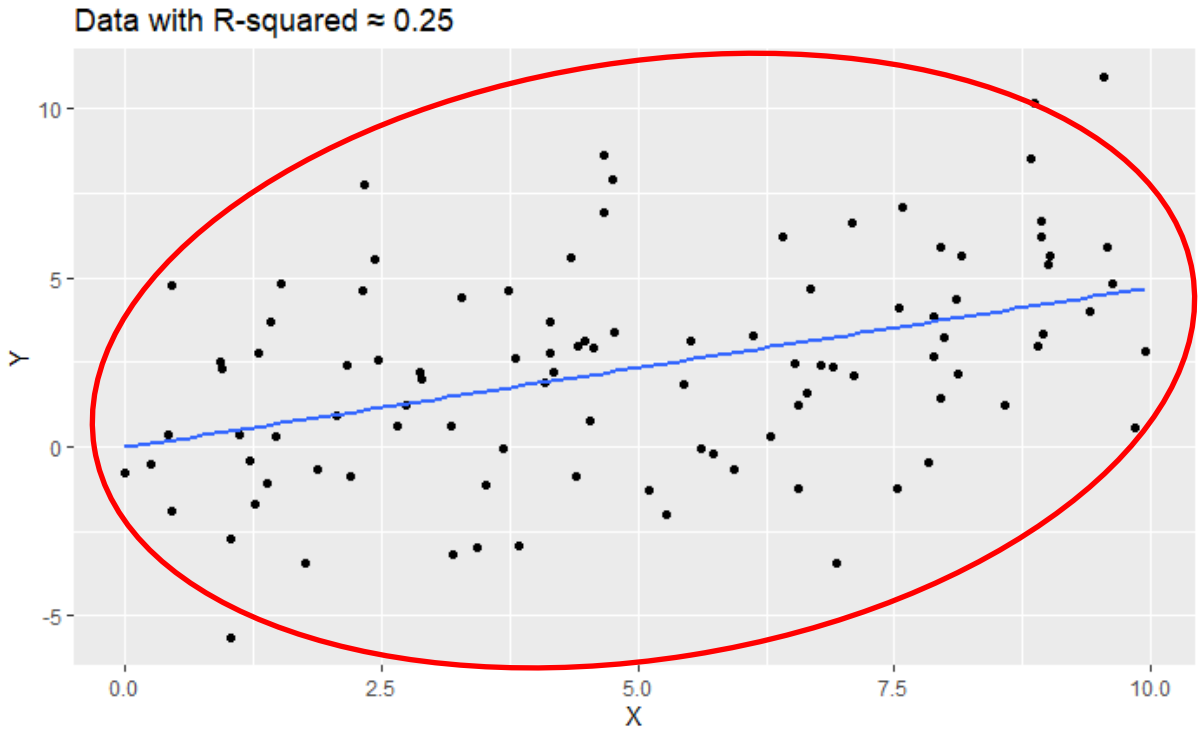
Model Error: Residuals

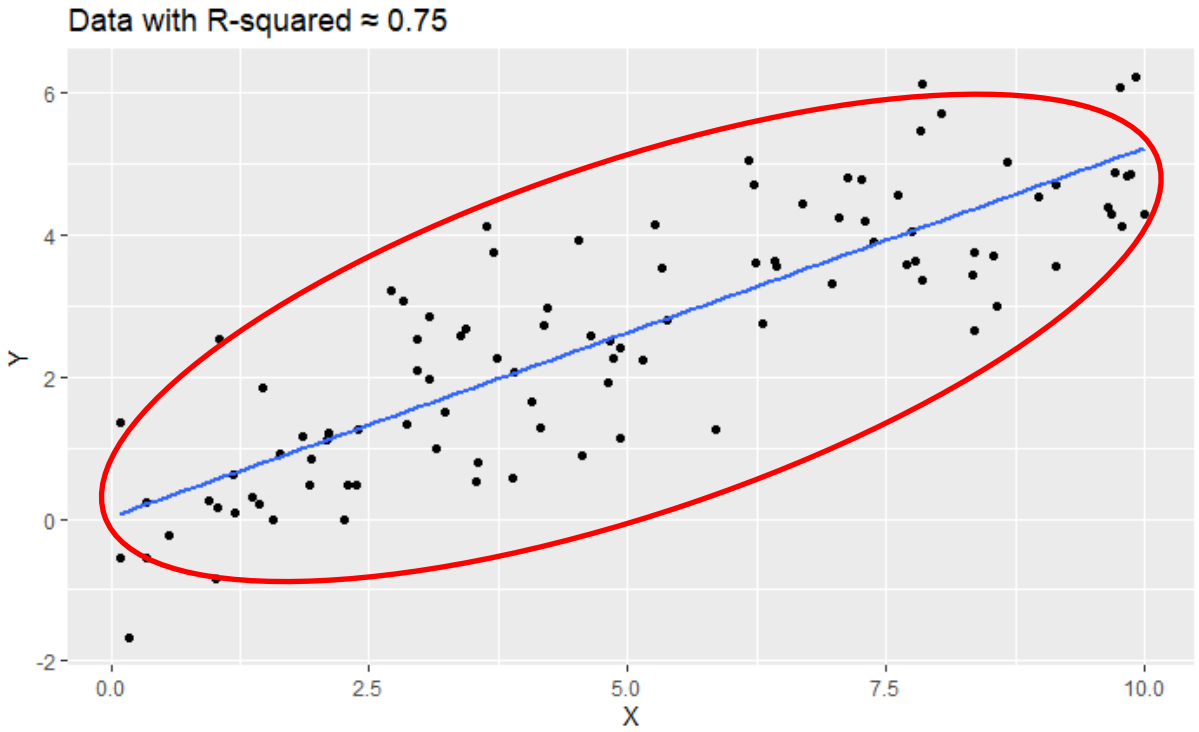


- **Residuals**: Change in Y between measured (red) and predicted (green) values
- Measured value = 50.8
- Predicted value = 63.2
- Residual = 50.8 – 63.2
- Residual = -12.4

- Model overestimates the measured value by 12.4

Model Error: R-Squared (R^2)





Model Error: Error Term (ϵ)

- Consider the following regression model
  - **Commute Time** (Y) = $\beta_0 + \beta_1$[**Distance** (X_1)]

- What other factors might influence commute time?

Model Error: Error Term (ϵ)

- Consider the following regression model
  - Commute Time (Y) = $\beta_0 + \beta_1$[Distance (X_1)] + ϵ
  - **ϵ (epsilon)**: the error term represents all variables not captured by the regression model

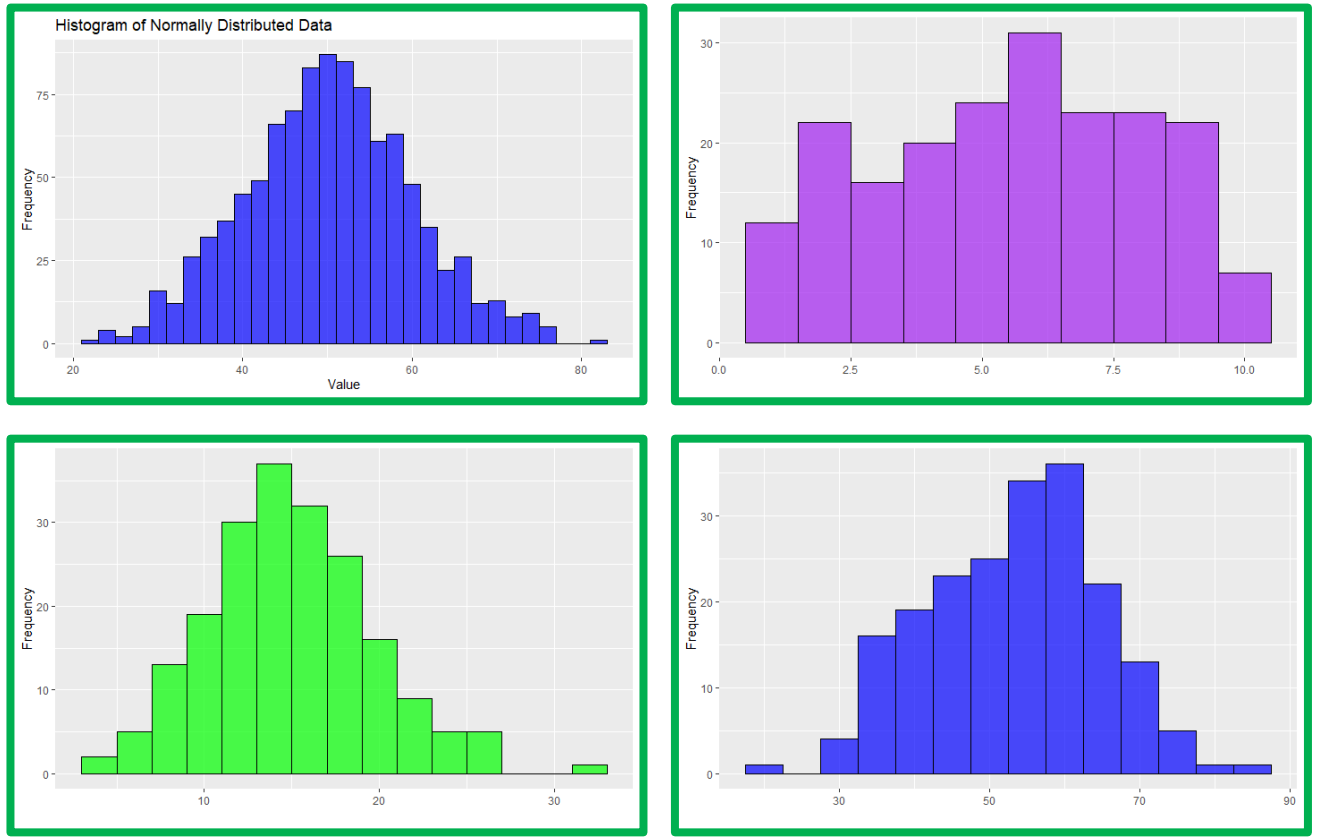
Multivariate Linear Regression

$$Y = \beta_0 + \underbrace{\beta_1 X_1}_{\text{Independent Variable 1}} + \underbrace{\beta_2 X_2}_{\text{Independent Variable 2}} + \underbrace{\beta_3 X_3}_{\text{Independent Variable 3}} + \ldots + \underbrace{\beta_n X_n}_{\text{Independent Variable n}} + \epsilon$$

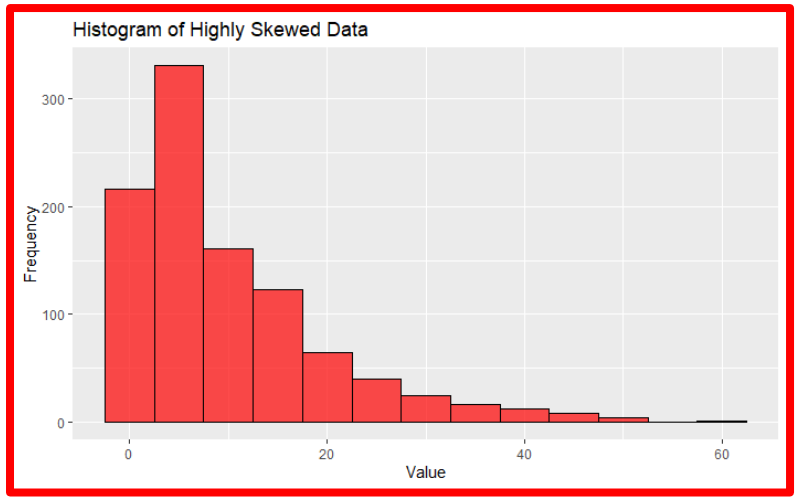
Model Error: Adjusted R-Squared (R^2)

Number of Observations (n)	Number of Variables in Regression Model	R-Squared (R^2)	Adjusted R-Squared (Adjusted R^2)	Comments
100	2	0.60	0.50	
100	3	0.70	0.65	
100	4	0.75	0.70	Ideal Number of Variables
100	5	0.80	0.68	Over-fitted model, additional variables not providing enough explanatory power
100	6	0.85	0.55	

Exploratory Data Analysis

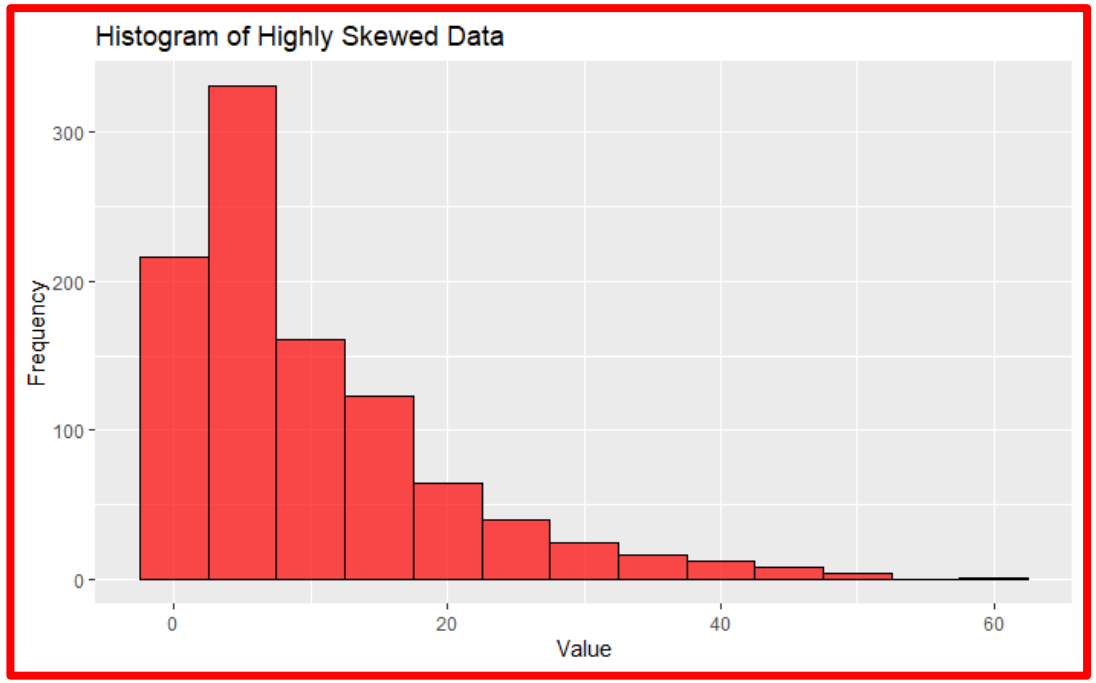


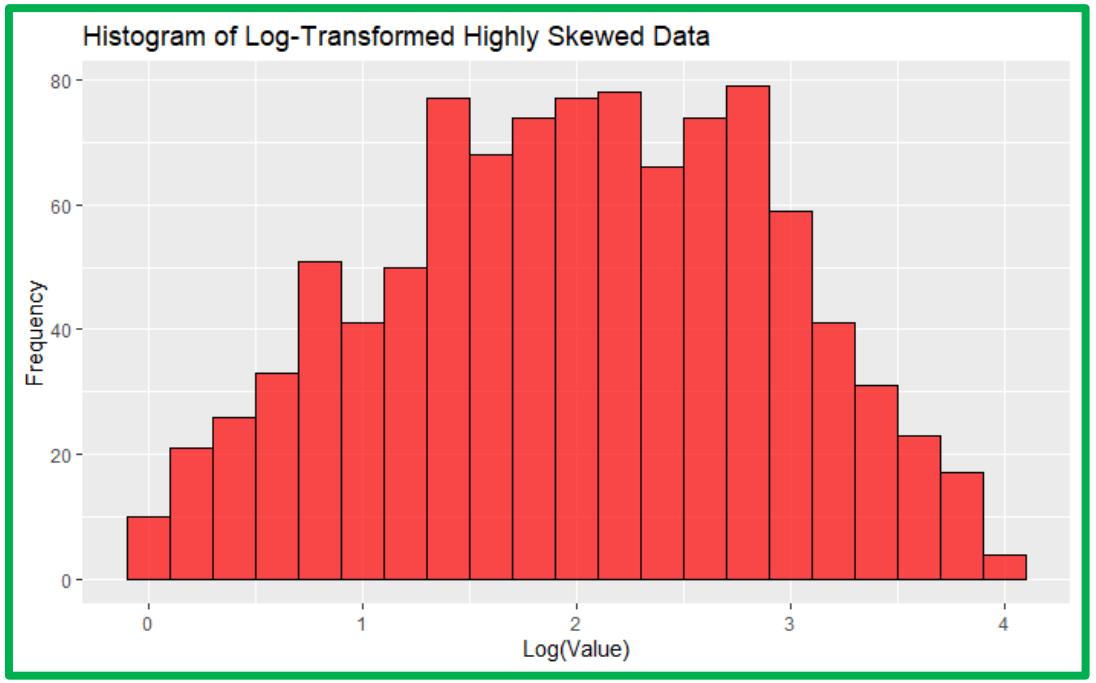
✅ Normal Distribution



❌ Highly Skewed Data

Log-Transforming Data





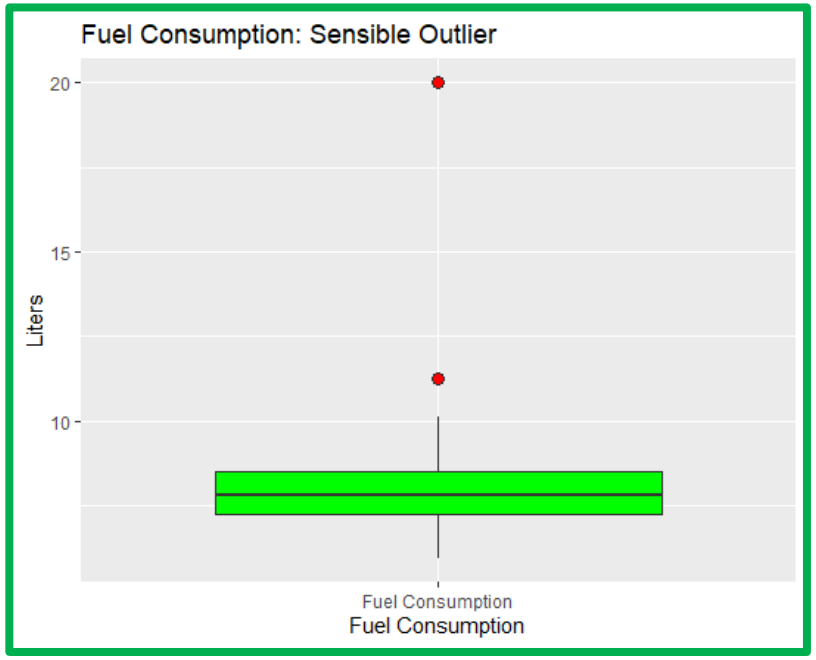
❌ Highly Skewed Data

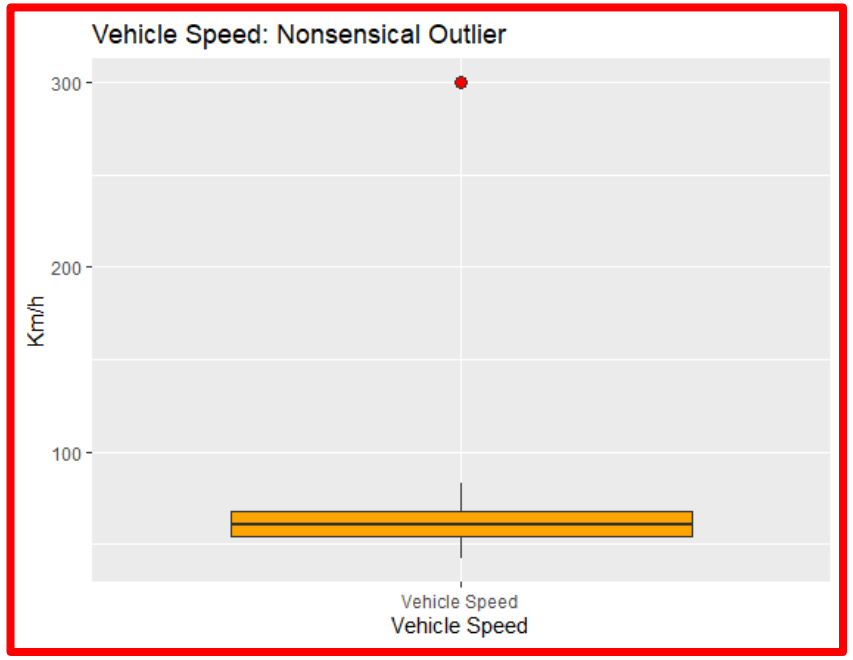
Untransformed data

✅ Normal Distribution

Log-Transformed Data

Exploratory Data Analysis





✅ No outliers

✅ Keep outliers

❌ Remove outlier

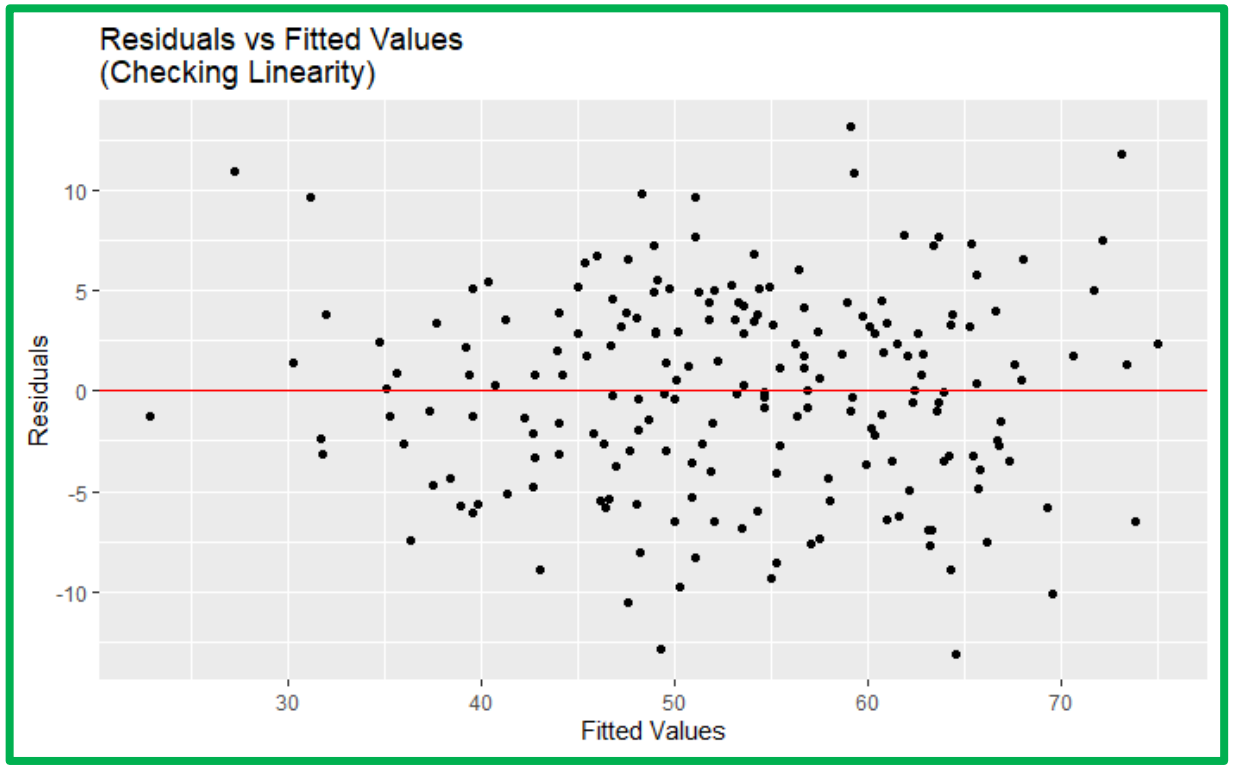
Null Hypothesis

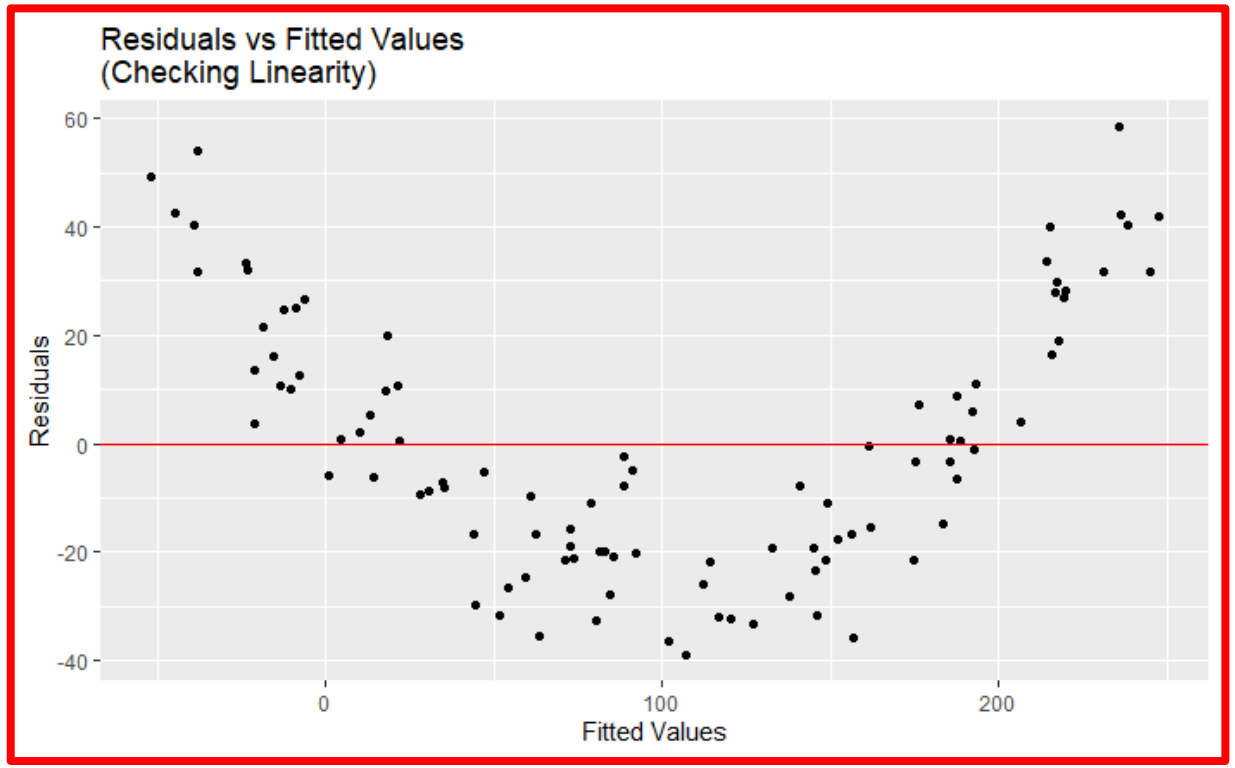
- **Null hypothesis**: no relationship between variables

- **p-value**: the probability of obtaining an observed result (or one more extreme) assuming that the null hypothesis is true

- **"Statistically Significant"**: Suggests that your result is unlikely to have occurred by random chance alone
  - **Good**: p-value <0.05
  - **Better**: p-value <0.01
  - **Best**: p-value <0.001

Assumptions of Linear Regression: Linearity





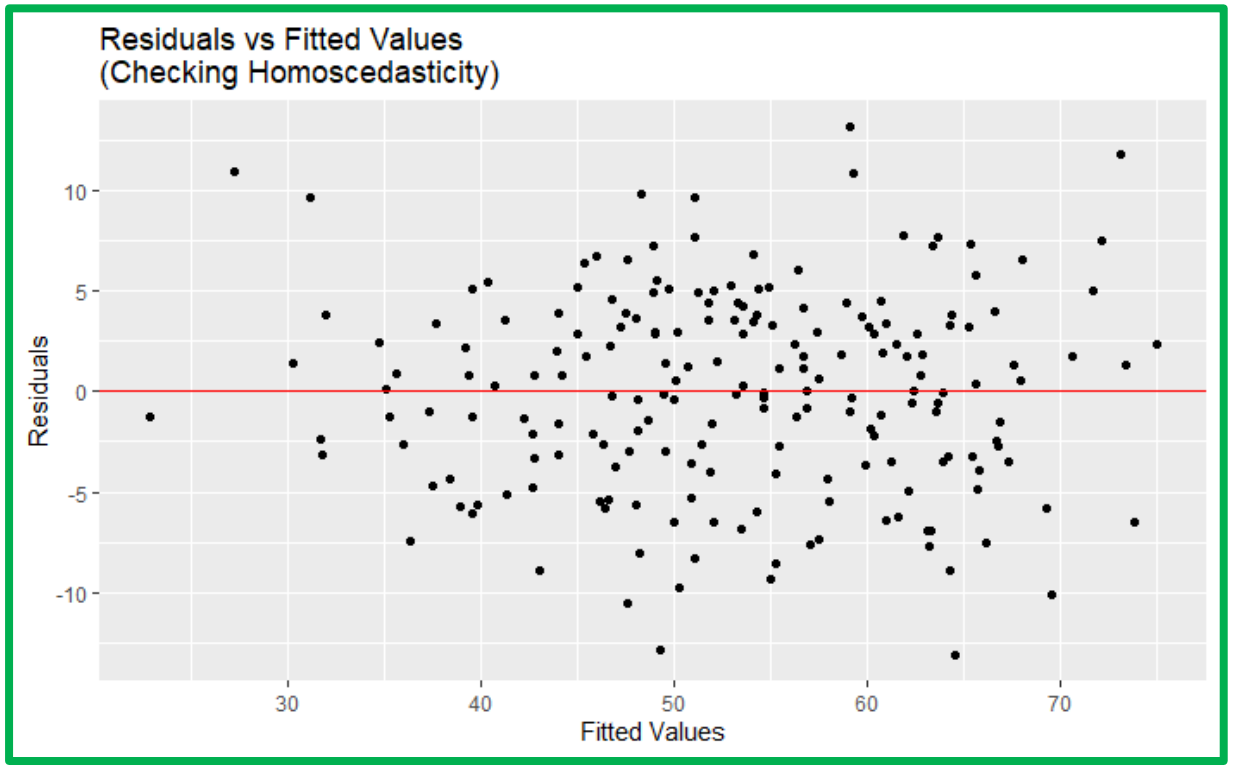
✅ Linear Relationship

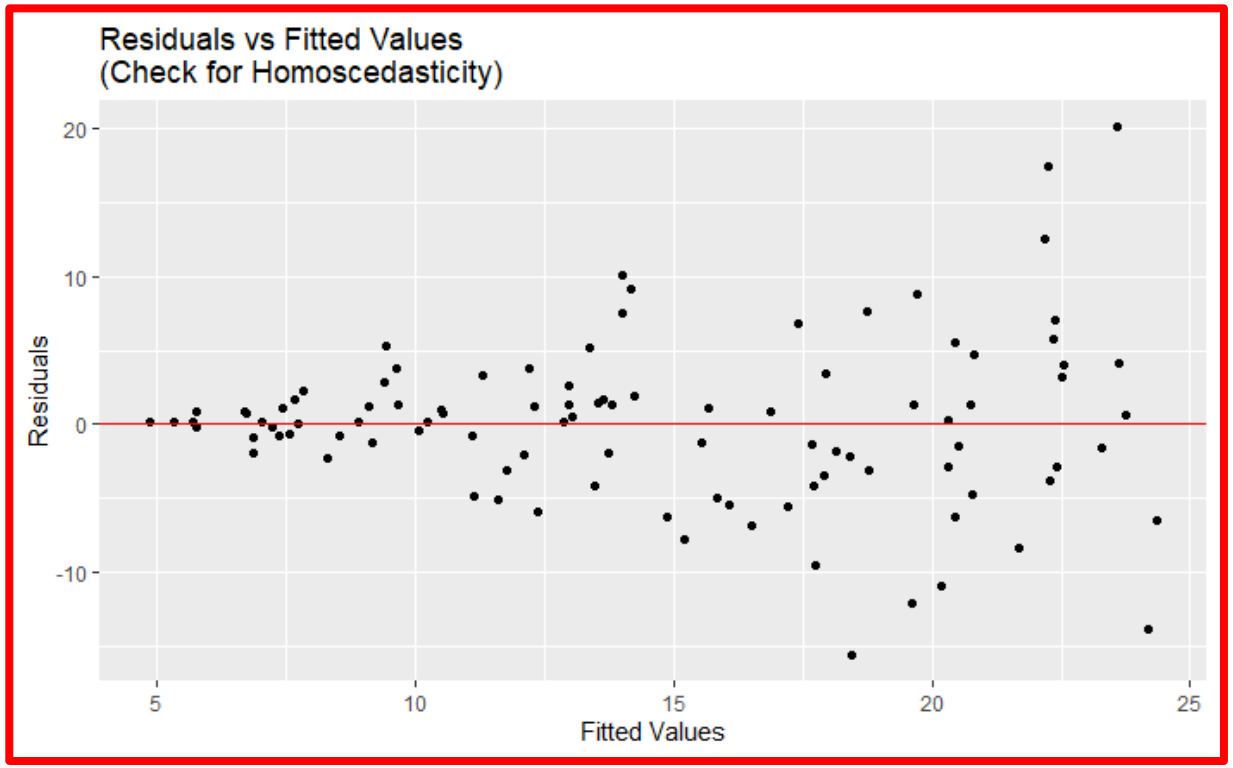
The relationship between X and Y is linear

❌ Non-Linear Relationship

Log-transform variables

Assumptions of Linear Regression: Homoscedasticity





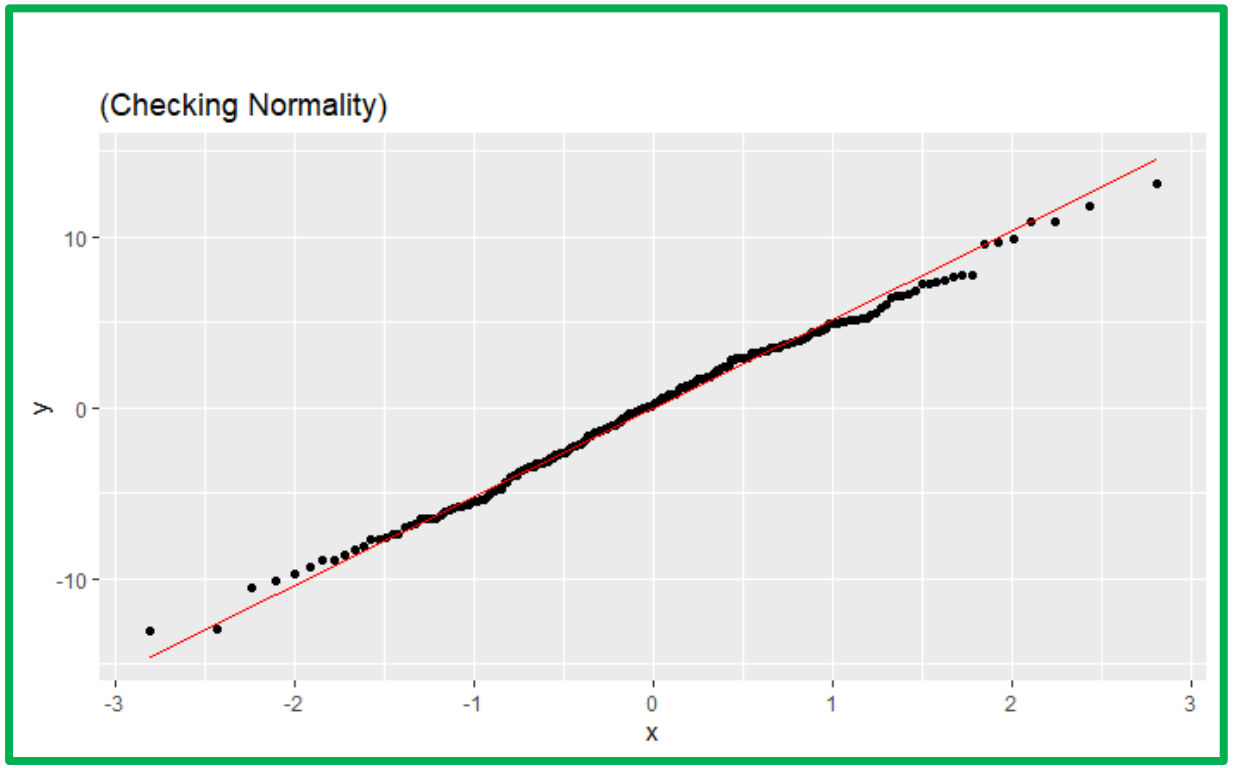
✅ Homoscedasticity

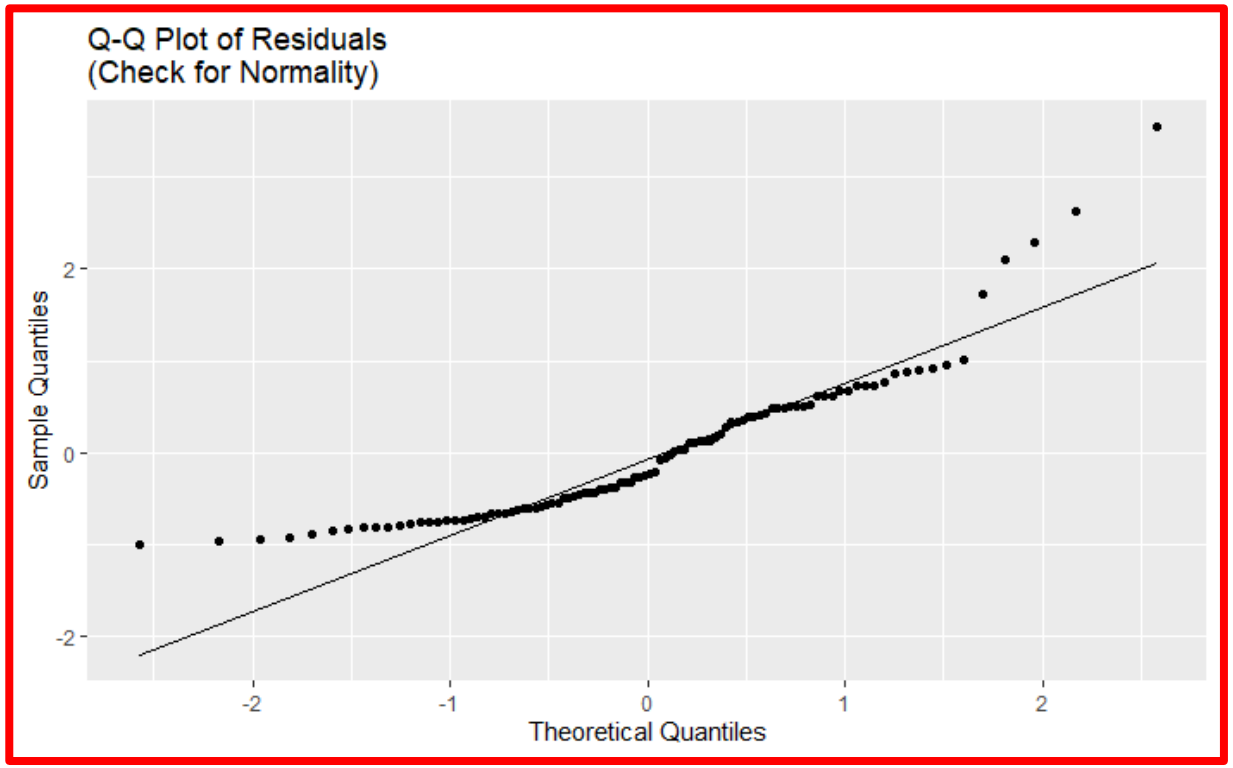
Constant variance of errors

❌ Heteroscedasticity

Log-transform dependent variable

Assumptions of Linear Regression: Normality of Residuals





✅ Normally Distributed

Errors are normally distributed

❌ Not Normally Distributed

Log-transform dependent variable

Assumptions of Linear Regression: Independence

Durbin-Watson test
DW = 2.1317, p-value = 0.8233

✅ Residual independence

Observations are independent of each other

Durbin-Watson test
DW = 0.4242, p-value = <0.001

❌ Residuals autocorrelated

Use different regression model

Assumptions of Linear Regression: Multicollinearity

Variance of Inflation Factors

Distance: 1.01
Traffic Level: 1.01

✅ No Multicollinearity

Independent variables are not highly correlated

Variance of Inflation Factors

Distance: 1.01
Traffic Level: 4.03
Traffic Density: 5.29

❌ Multicollinearity

Remove variable with highest variance of inflation factor

Interpreting A Linear Regression Model

- Source: (Table 2) Effects of street pattern, traffic, road infrastructure, socioeconomic and demographic characteristics on public transit ridership
- **Dependent Variable**: Transit Ridership (%)
  - Percent of total commuter trips as public transit
- **Independent Variables**: 22 total variables
  - Unemployment Rate (%)
  - Total Commercial Area (km^2)

Interpreting The Results: Transit Ridership (%)

Coefficients	Estimate	p-value	
Constant / Intercept (β_0)	-16.181	0.045	*
% Unemployment Rate (β_1)	0.881	0.001	***
Total Commercial Area, km^2 (β_2)	0.038	0.064	
19 Other Variables (β_3 to β_{22})	See paper	See paper	
R-Squared: 0.68			
Adjusted R-Squared: 0.64			

Thank You For Listening!

Dr. Kevin Mackay
Data Analyst, Statistics Canada